DATA ANALYTICS PROJECT

DATA ANALYTIC & VISUALIZATION (1IT508)

HEART DISEASE

From raw records to a clear Power BI dashboard

ENROLLMENT NO.: 240840116083
240840116084
240840116087
240840116128
240840116131

PROBLEM STATEMENT

Raw heart-health data may contain repeated records and is harder to interpret without cleaning and visual analysis.

Observation of the raw dataset

Source file: 16_ heart.csv • Initial inspection before cleaning

ROWS	COLUMNS	DUPLICATE ROWS	MISSING VALUES
1,025	14	723	0

Columns available

age • sex • cp • trestbps • chol • fbs • restecg • thalach • exang • oldpeak • slope • ca • thal • target

The target column is the outcome label; confirm its coding with the dataset documentation.

Main issue found

723 completely duplicated rows were detected (70.5% of the raw file).

No missing values were detected in the raw file.
Duplicate records need review before removal.

Data cleaning process

Cleaning was performed in Python using Pandas, then exported as cleaned_heart.csv.

01	**Load CSV**	Read the raw file into a Pandas DataFrame.
02	**Inspect**	Check shape, column types, missing values and duplicates.
03	**Remove exact duplicates**	Keep one copy of each fully identical row.
04	**Validate numeric fields**	Check age and oldpeak numeric types; inspect for conversion errors.
05	**Export**	Save the cleaned result to a new CSV, preserving the raw source.

Important: removing duplicates assumes repeated rows are unwanted copies. Verify this decision for any medical or patient-level analysis.

Solution: final clean dataset

Comparison of raw and cleaned files after exact duplicate removal

Metric	Raw data	Clean data
Rows	1,025	302
Columns	14	14
Exact duplicate rows	723	0

Final file

cleaned_heart.csv

302 rows × 14 columns

0 missing values • 0 duplicate rows

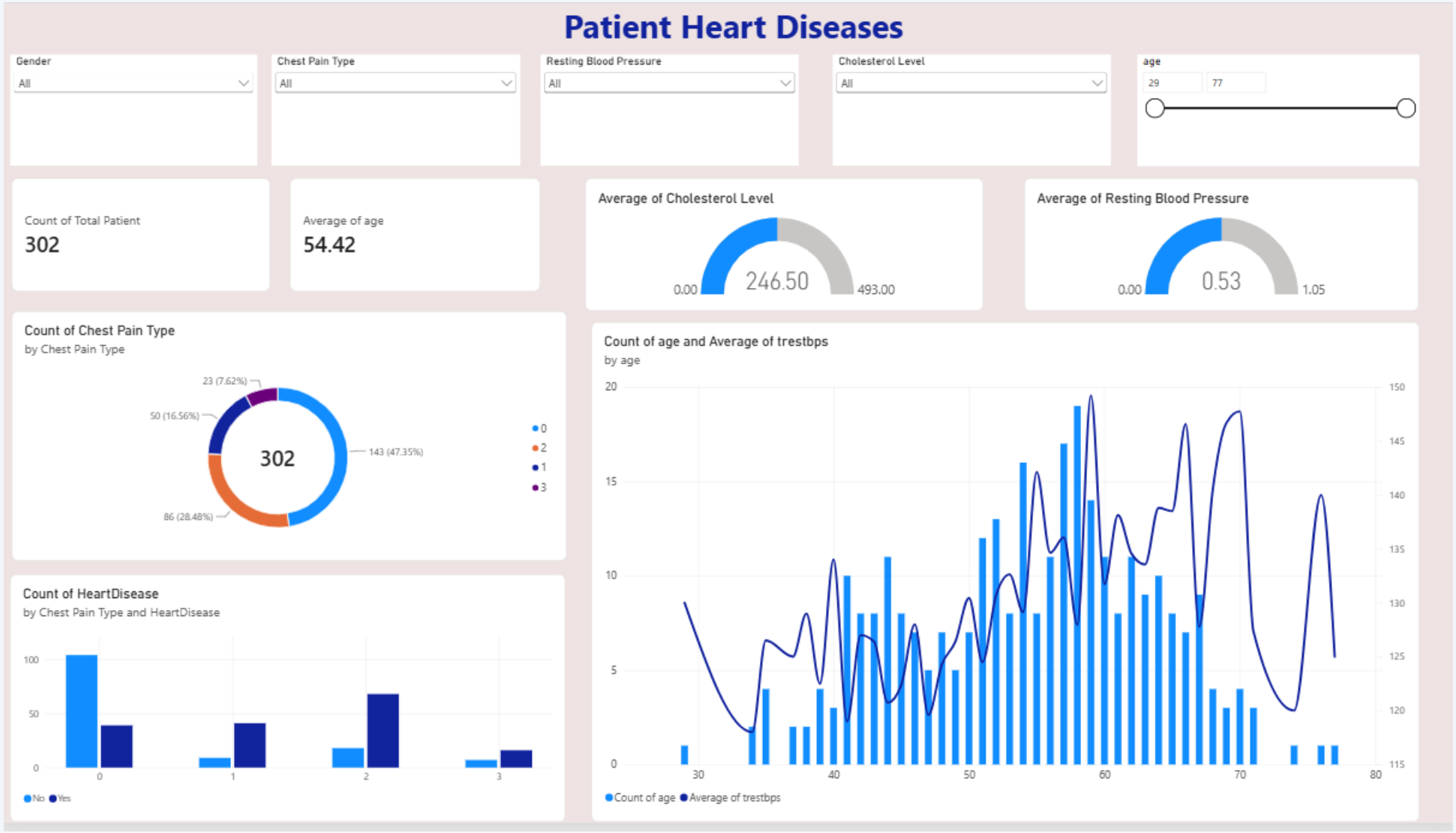
What improved?

Removed 723 exact duplicate rows (1,025 → 302). All 14 columns were retained. No missing values were present in either file.

No additional rows were dropped for outliers; unusual values should be reviewed with domain context rather than deleted automatically.

Visualizations of the cleaned dataset

A dashboard-style overview created from the cleaned heart dataset



Observations from the dashboard

Descriptive findings from the 302 unique records (not medical conclusions)

DISEASE PATIENTS	164 (54.3%)
NO DISEASE	138 (45.7%)
AVERAGE AGE	54.4 years
AVERAGE CHOLESTEROL	246.5

Key observations

- Mean resting blood pressure: 131.6; mean maximum heart rate: 149.6.
- Outcome groups: 164 records are target=1 and 138 are target=0 (assuming documented coding is 1=present, 0=absent).
- Chest-pain codes differ in how many disease-positive and disease-negative records they contain; interpret codes using the dataset documentation.

The target column is an encoded outcome. Results describe this dataset only and should not be used for diagnosis.

Recommendations and suggestions

Practical improvements for a reliable and useful Power BI report

01

Validate duplicate policy

Confirm whether identical rows are repeated copies or legitimate repeated observations before excluding them.

02

Verify category definitions

Document the meaning of sex, chest-pain, restecg, thal and target codes.

03

Review unusual values

Investigate high cholesterol or other unusual measurements; do not remove outliers without justification.

04

Improve dashboard usability

Use dropdown slicers for categories, range sliders for numeric fields, clear titles and consistent colors.

05

Add useful comparisons

Show disease-status counts by age group and chest-pain category, plus average cholesterol and blood pressure.

06

Refresh and document

Keep raw and cleaned files separate, record cleaning steps, and refresh the report when new data arrives.

CONCLUSION • Cleaning improves consistency; visualization makes patterns easier to explore.